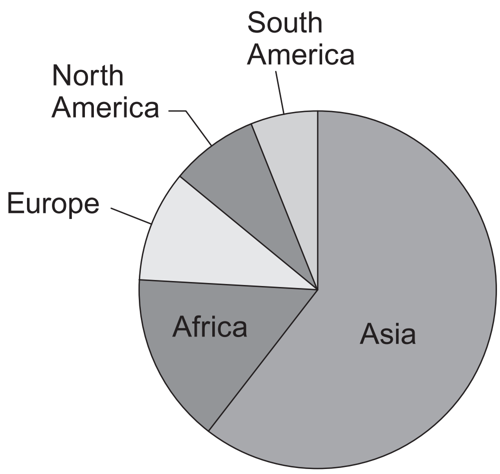
5. Plant and crop growth can be improved using fertilisers containing nitrogen. Fertilisers are substances that are added to the soil in order to increase the supply of nutrients that boost the growth of plants. With the rapid increase in global population, the demand for food has been rising tremendously. It is estimated that 40-60% of agricultural crops are now grown with the use of different types of fertilisers. Many fertilisers are produced using ammonia.

When you use too much fertiliser in the soil, it can lead to eutrophication. Fertilisers contain substances like nitrates and phosphates that are washed into lakes, oceans and rivers by rain water. These substances lead to excessive growth of algae and plants in the waterways resulting in a decrease in the levels of oxygen. The decrease in oxygen levels leads to the death of fish and other aquatic animals and contributes to changes in food chains.

Figure 1 shows the relative populations of world continents in 2016.

Figure 2 shows the relative world consumption of ammonia in 2016.

Figure 1



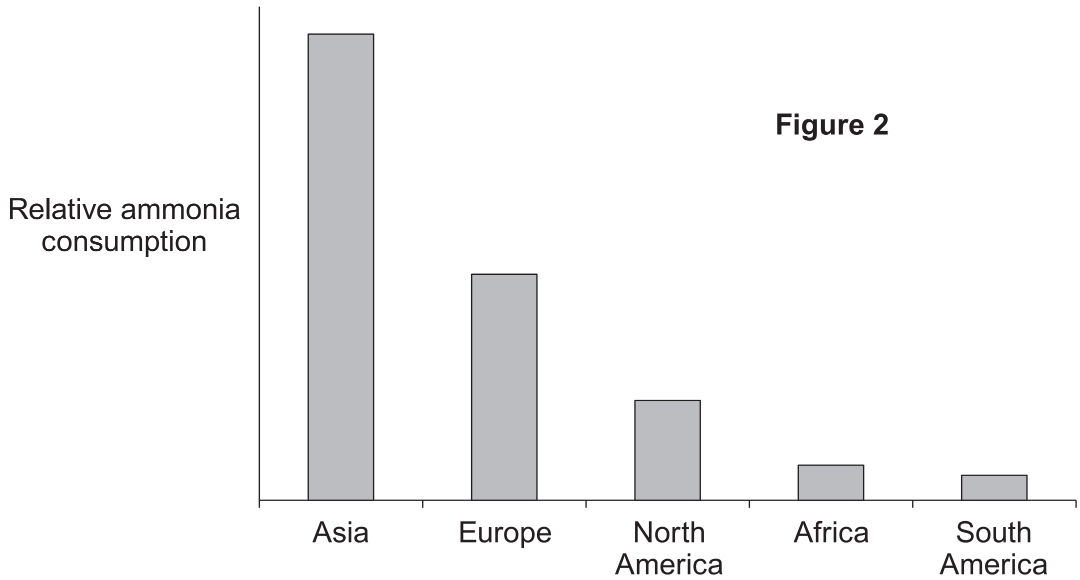


Figure 2

Examiner only

(a) (i) Tick (✓) **three** boxes that state the effects of fertiliser use which contribute to the death of aquatic animals in eutrophication. [2]

crop growth on fields increases	
fertilisers run into waterways	
plant growth in rivers and lakes increases	
aquatic animals do not have enough oxygen	
farmers' profits increase	

(ii) State whether the information in **Figure 1** and **Figure 2** shows a link between the number of people living on each continent and the amount of ammonia used. Give reasons for your answer. [2]

....

....

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Examiner only

(b) The label on a bag of fertiliser shows the formula of the nitrogen compound that the fertiliser contains.

1.5 kg fertiliser

(i) Give the name of the compound $(NH_4)_2SO_4$. [1]

....

(ii) Tick (✓) the box that gives the correct reason why plants need nitrogen. [1]

plants use nitrogen to make sugar	
plants use nitrogen to make water	
plants use nitrogen to make oxygen	
plants use nitrogen to make protein	

(iii) 1.5 kg of fertiliser treats an area of $75\,m^2$. Calculate the mass needed to treat a lawn with an area of $15\,m^2$. Give your answer in **grams**. [2]

Mass = ... g

Examiner only

(c) Ammonia is used to make many fertilisers. Ammonia gas is produced from nitrogen and hydrogen using the Haber process.

(i) Balance the equation for the reaction between nitrogen and hydrogen in the Haber process. [1]

$$N_2 + \boxed{}\,H_2 \rightleftharpoons \boxed{}\,NH_3$$

(ii) State what is meant by the symbol $\rightleftharpoons$ used in the reaction equation. [1]

....

(iii) A catalyst is used in the Haber process.

I. Give the name of the catalyst used. [1]

....

II. State why a catalyst is used. [1]

....

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